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Q-1 Step - 1 normalize RGB

R = 30/255 = 0.118

G = 108/255 = 0.424

B = 210/255 = 0.824

Step - 2 CMY conversion

CMY = 1 - RGB

C = 1-0.118 = 0.882

M = 1 - 0.424 = 0.576

Y = 1-0.824 = 0.176

Step - 3 HSI conversion

Intensity (I)

I = (R+G+B)/3 = 0.455

Saturation (S)

min RGB = 0.118

S = 1 - (min/I) = 0.741

Hue (H)

Computed angle ≈ 214.4°

Step 4 : 3D HSI

H = 214.4° → angle

S = 0.741 → radius

I = 0.455 → height

Cartesian form :

x = -0.612

y = -0.419

z = 0.455

Q-2 Approach

1. Identify intensity range of soft tissue
2. Assign a specific color to that slice
3. Assign another color to bone intensities
4. Keep background dark

Result:
- Bones appear in a bright contrasting color
- soft-tissue areas get a different color band.
- subtle differences become visually separable.

Q-3 1. Smoothing reduces high-frequency noise and preserves broad structure
2. Pseudo-color mapping assign colors to intensity bands.

When the intensity landscape becomes smoother, the color transitions reveal:
- Previously invisible gradients.
- hidden weak edges.
- Small but real intensity variations.

Q-4 Different tissues have different grayscale intensities.
fat - low intensities
muscle - medium
Bone - high

Pseudo-color mapping assign distinct colors to each intensity zone.
fat → blue
muscle → yellow
bone → ~~red~~ red

Q-5 FCC replaces the visible R-G-B channel with infrared and other wavelengths.
NIR → Red
Red → Green
Green → Blue

This helps.
Vegetation appears bright, red
water bodies look dark
bare soil shows brown or yellow
urban regions appear cyan or gray.

Q-6 Memory, Blue-ray requirement
Resolution = 1084 x 720
Frame Rate = 80 fps.
Duration = 3 hours = 3 x 3600 = 10800 s.
Color depth = 24 bits/pixel.

a) Total uncompressed memory
pixel per frame
= 1084 x 720 = 780,480 pixels.
Bits per frame.
= 780,480 x 24 = 18,731,520 bits.
= 2.34 MB/frame.
Num of frame.
= 80 x 10800 = 864,000 frames.
Total size:
2.34 MB x 864,000
≈ 2,022,000 MB

b) Num of 50 GB Blu-ray discs
2,022 GB / 50 GB = 40.4 discs.

c) Transmission Time at 100 mbps
total bits = 2.02 TB
= 2.02 x 8 x 10^{12} bits
= 16.16 x 10^{12} bits
Time = (16.16 x 10^{12}) / (100 x 10^{6})
= 161600 s
= 44.8 hours

Q-7 Step : 1 Count frequencies

Values and counts

1 → 5
2 → 5
8 → 3
9 → 3
7 → 2
6 → 3
3 → 2
5 → 2
0 → 1

Step - 2 Build Huffman Tree

1 → 00
2 → 01
8 → 100
6 → 101
9 → 110
7 → 1110
3 → 11110
5 → 11111
0 → 111010

Step - 3 Average code length

Average = 3.1 bits / symbol.

Step - 4 Total bits required

25 pixel × 3.1 bits = 77.5 bits

≈ 78 bits

Step 5 : Entropy

Compute using $-\sum p \log_2 p$

Entropy ≈ 3.07 bits / symbol

Step - 6 Compression ratio

Original 8 bits x 25 = 200 bits.

Compressed : 78 bits,

Compression ratio = 200/78 ≈ 2.56 : 1

Q-8 Why DSPs are preferred over FPGA & ASIC in embedded systems

DSPs win in real products because :

- They support fast multiply-accumulate operations
- have optimized instruction sets for filtering, transforms control loops.
- Provide low power consumption compared to FPGA.
- Easier programming compared to HDL for FPGA.
- Lower cost and faster development vs ASIC.
- Include timers, ADC / DAC interfaces, PWM for real for motors and audio

Q-9 Typical DSP pipeline

1) Fetch - Instruction loaded from program memory
2) Decode - Identifies operation and reads registers
3) Address Generate - memory address computed.
4) Execute - MAC operation happens here
5) Write Back - store results in registers

Q-10. Key Features

- MAC unit
- Harvard architecture
- Circular buffers
- pipelining.
- zero - overhead loops
- Special addressing modes.